

Big Data and Data Visualization



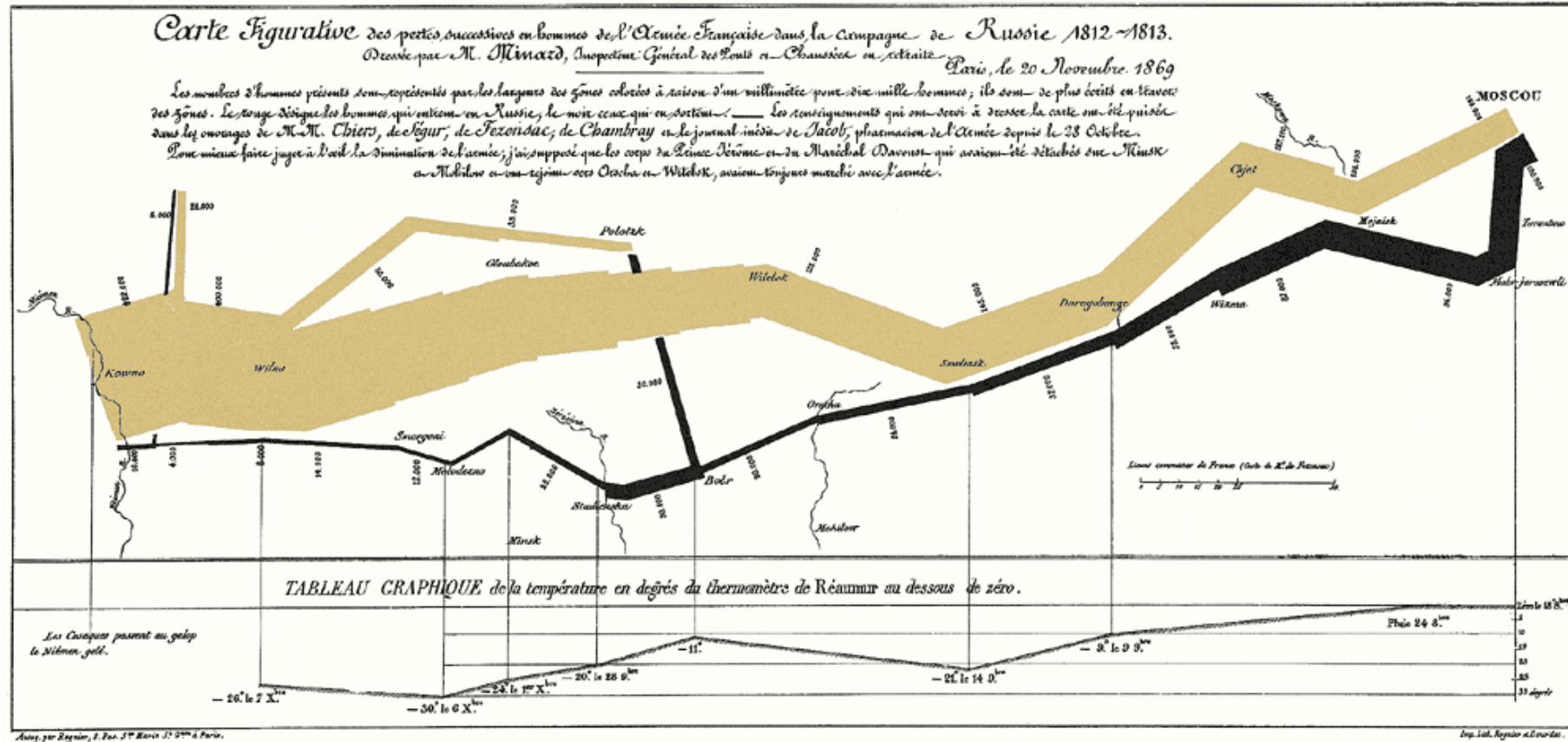
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A Brief History of Data Visualization

What makes a good chart?

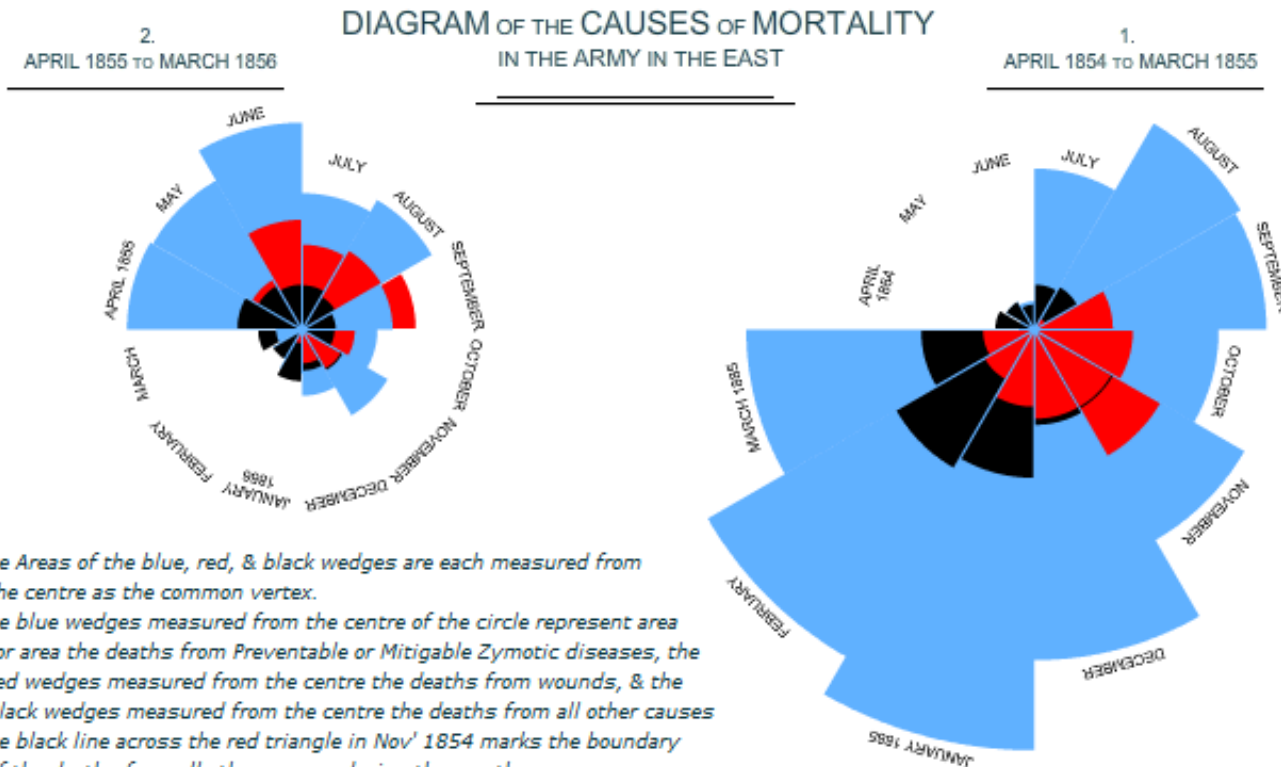


Napoleon's 1812 March by
Charles Joseph Minard

Perhaps the most famous data presentation...

Reprinted in Tufte (2009), p. 41

Florence Nightingale's 'Coxcombs' 1858



The Areas of the blue, red, & black wedges are each measured from the centre as the common vertex.

The blue wedges measured from the centre of the circle represent area for area the deaths from Preventable or Mitigable Zymotic diseases, the red wedges measured from the centre the deaths from wounds, & the black wedges measured from the centre the deaths from all other causes

The black line across the red triangle in Nov' 1854 marks the boundary of the deaths from all other causes during the month

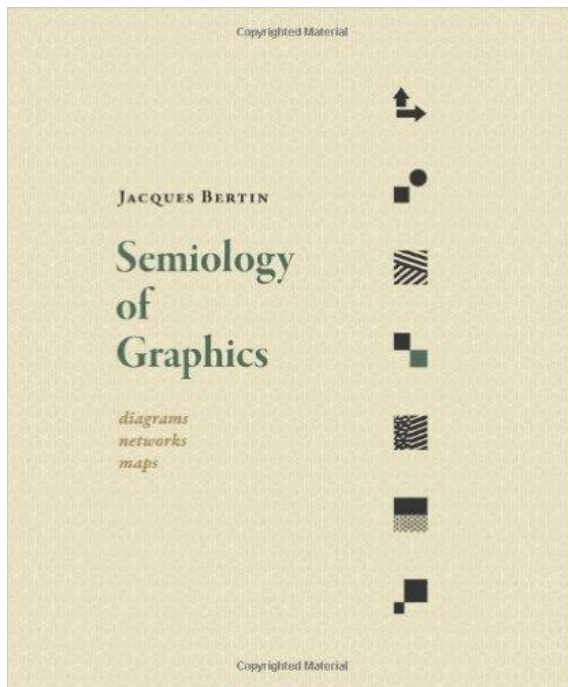
In October 1854, & April 1855, the black area coincides with the red, in January & February 1856 the blue coincides with the black

The entire areas may be compared by following the blue, the red & the black enclosing lines.

- Pioneer hospital sanitation
- Meticulously gathered data
- Pioneer in applied statistics and visualization
- Nurse

Jacques Bertin

Seven Visual Variables

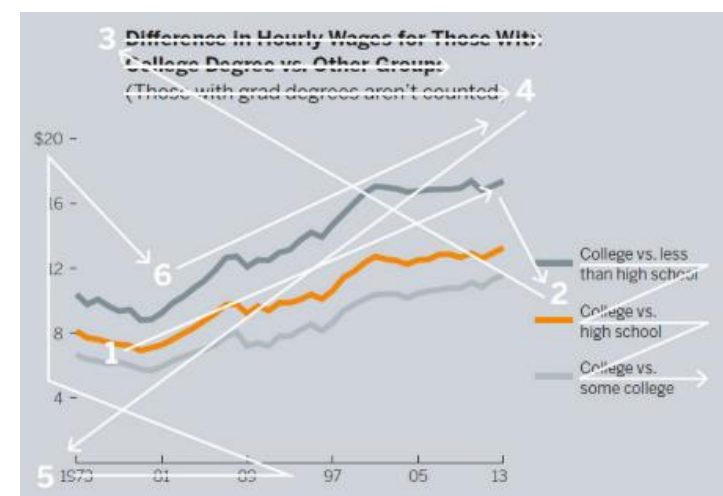
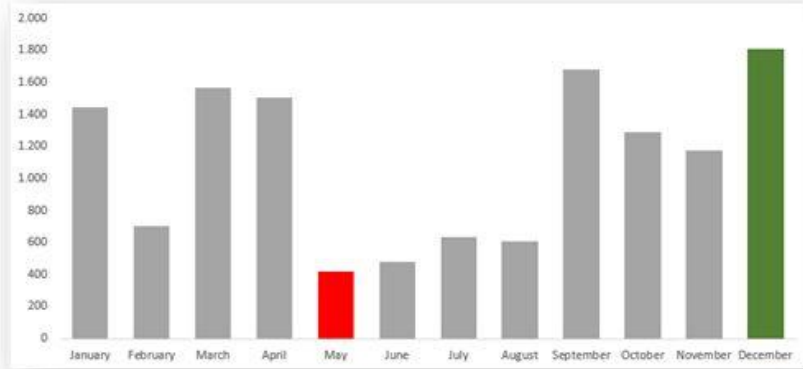


Bertin's Original Visual Variables	
Position changes in the x, y location	
Size change in length, area or repetition	
Shape infinite number of shapes	
Value changes from light to dark	
Colour changes in hue at a given value	
Orientation changes in alignment	
Texture variation in 'grain'	

- Position
- Size
- Shape
- Color
- Brightness
- Orientation
- Texture



When a Chart hits our Eyes



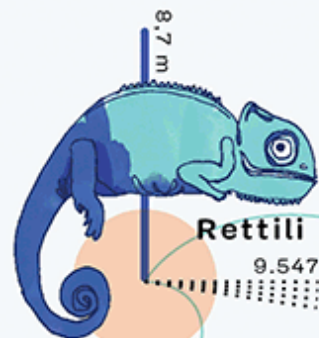
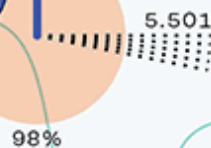
1. Visuals aren't read in a predictable, linear way
 - Create charts spatially, from the visual outward
2. We see first what stands out
 - Whatever stands out should support idea
3. We see only a few visuals at once
 - Plot as few visual elements as possible
4. We seek meaning and make connection
 - Relate visual elements in a meaningful way
5. We rely on conventions and metaphors
 - Embrace deeply ingrained conventions



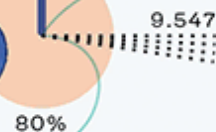
Gli animali formano un regno immenso



Mammiferi



Rettili



ognuna
con le proprie
caratteristiche,



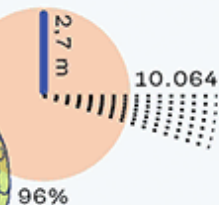
Anfibi



e composto
da così tante
specie.



Uccelli



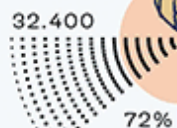
massima tra gli animali
della categoria

Numero
di specie
scoperte

Percentuale
di specie scoperte
sul numero di specie
stimate



Pesci



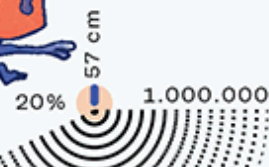
Molluschi



Pare che esistano
20 miliardi di miliardi
di animali nel mondo!



Insetti

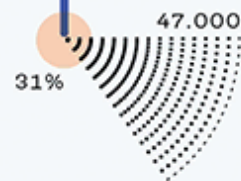


che gli uomini
non hanno ancora
trovato il tempo

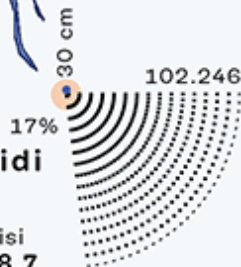
Crostacei



di classificarli
tutti!



Aracnidi



Suddivisi
in ben 8,7
milioni
di specie
diverse!

What is data visualization and why is it important?

- **Data visualization** is the act of taking information (**data**) and placing it into a visual context, such as a map or graph. **Data visualizations** make big and small **data** easier for the human brain to understand, and **visualization** also makes it easier to detect patterns, trends, and outliers in groups of **data**.

Is data visualization a part of data science?

- **Data science** and **data visualization** are not two different entities. They are bound to each other. **Data visualization** is a **subset of data science**. **Data science** is not a single process or a method or any workflow.

What is data discovery and visualization?

- **Data discovery** is the process of breaking complex **data** collections into information that users can understand and manage. It turns incomprehensible mounds of raw **data** into groups, sets, and relationships, making order out of chaos.
- **Data discovery** answers the question, “What does it all mean?”
- Data visualization is its representation.

What are data visualization tools?

- By using visual elements like charts, graphs, and maps, **data visualization tools** provide an accessible way to see and understand trends, outliers, and patterns in **data**.

Is Excel a data visualization tool?

- **Excel** is a spreadsheet **tool**, while Tableau is a **data visualization** one.
- Spreadsheet **tools** are electronic worksheets that display **data** in a tabular format (a table of columns and rows).
- Each **data** point is stored in “cells” and can be manipulated by manually set formulas.

How do you create good data visualization?

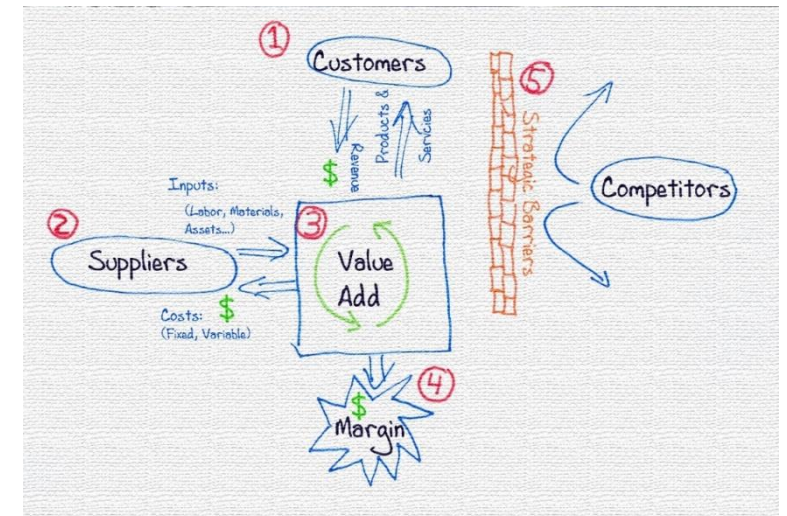
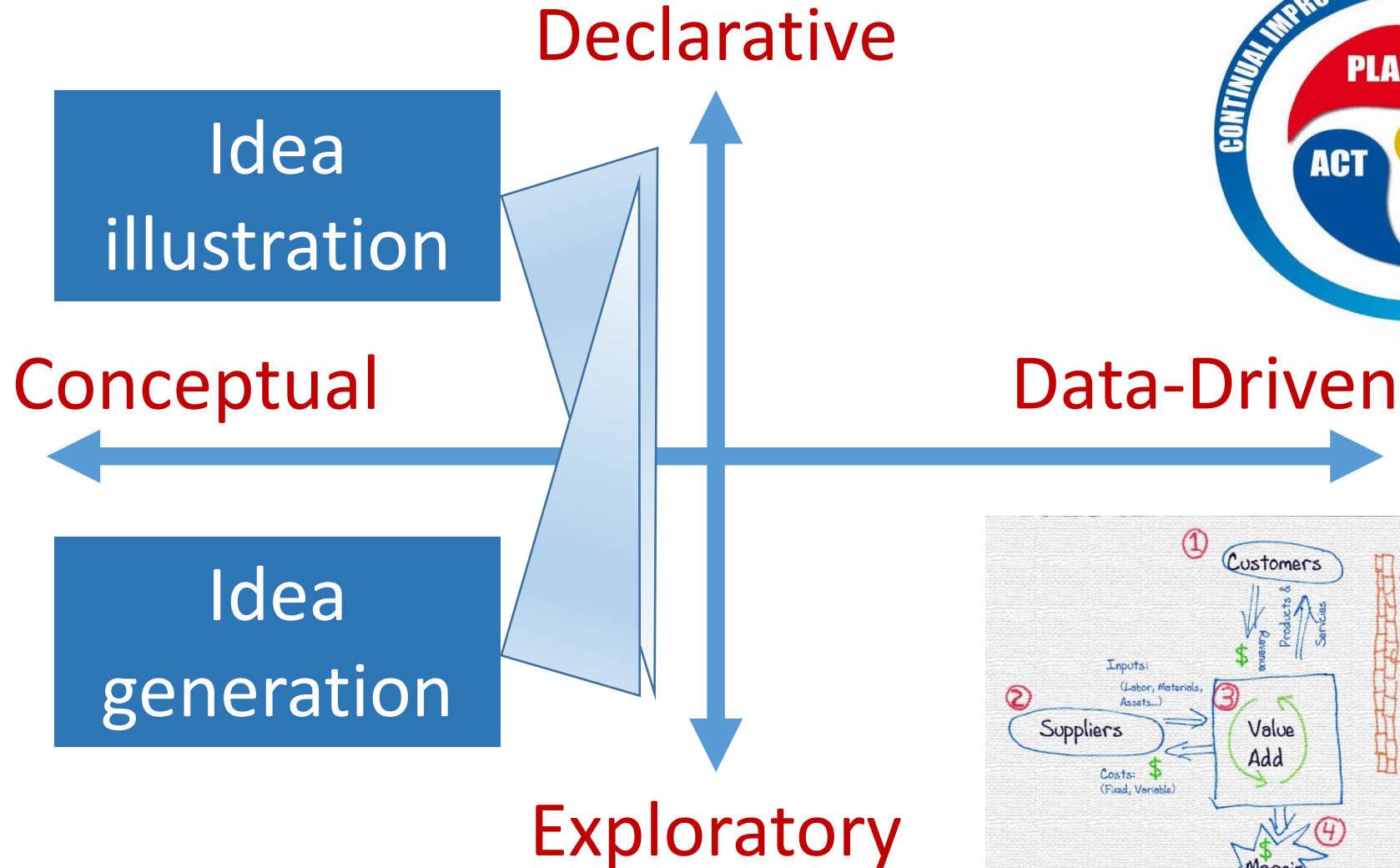
- Use it wisely in your data visualization design.
- Use a single color to represent the same type of data.
- Watch out for positive and negative numbers.
- Make sure there is sufficient contrast between colors.
- Avoid patterns.
- Select colors appropriately.
- Don't use more than 6 colors in a single layout.

What kind of visual communication do you want to create?

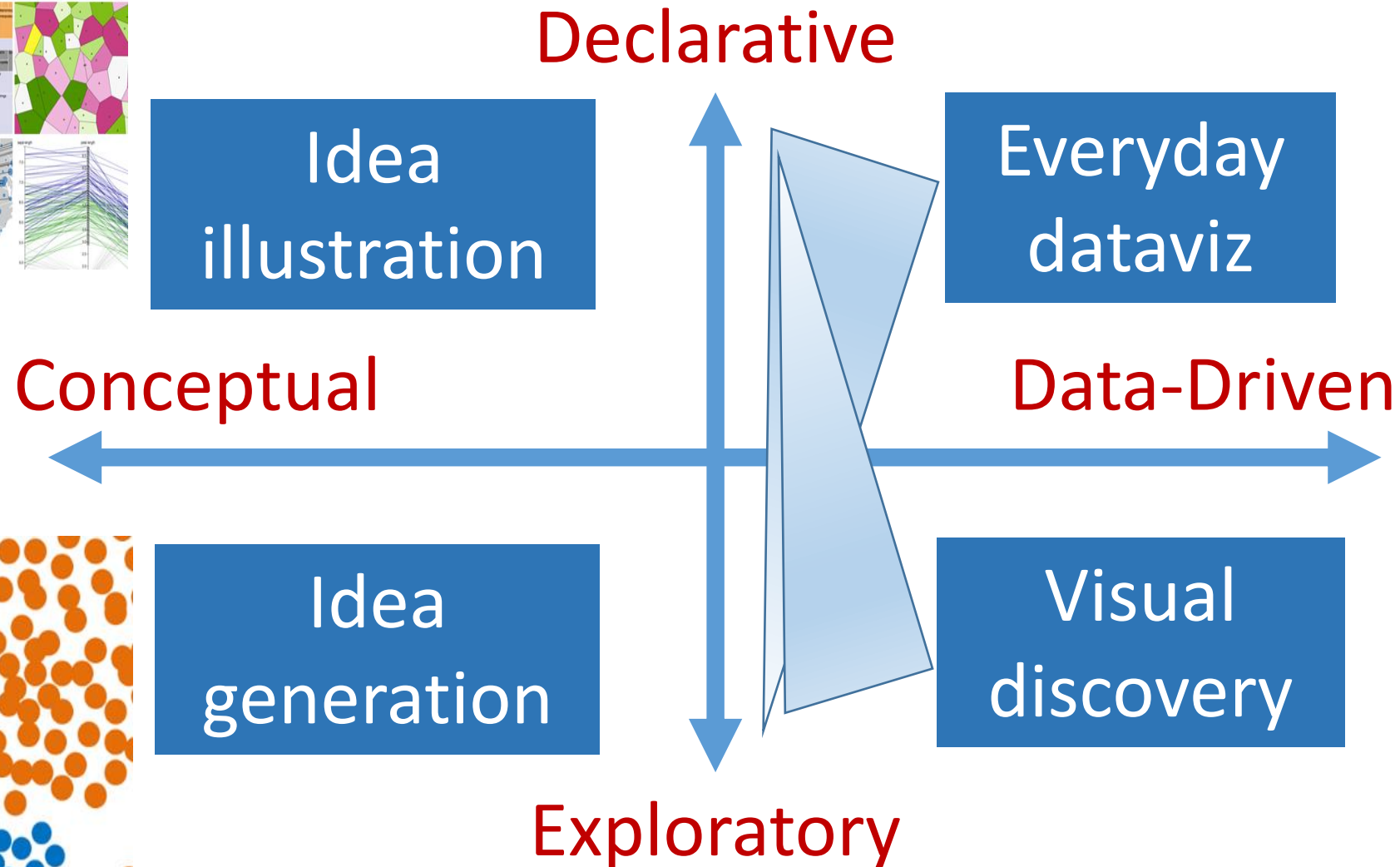
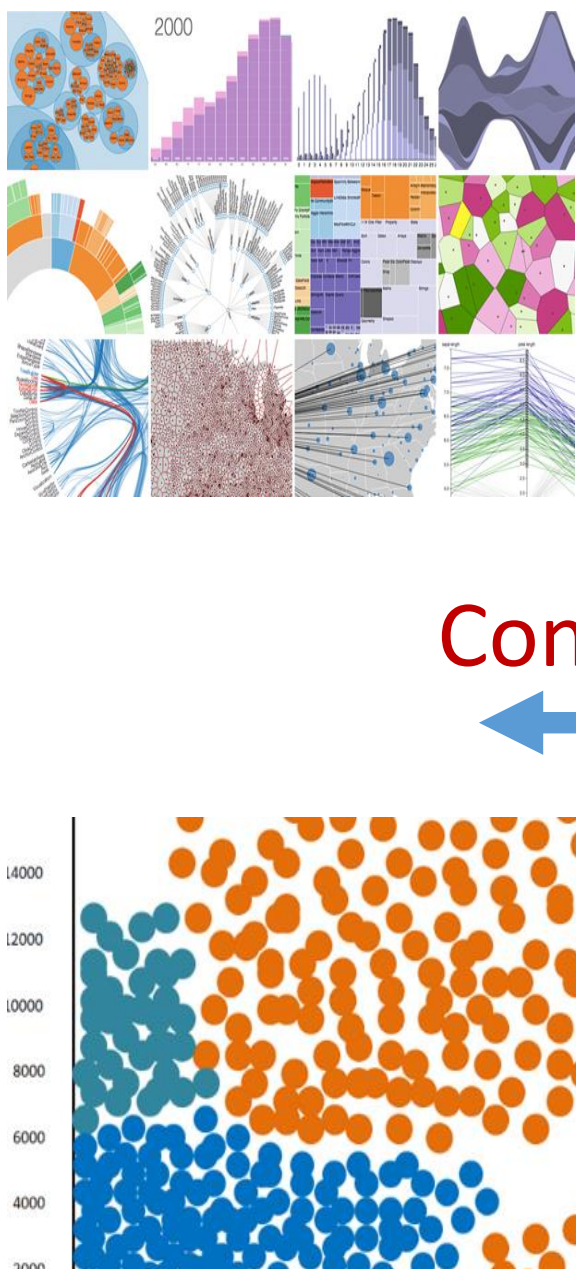


1. Is my information conceptual or data-driven?
 - Conceptual information is qualitative
 - Data-driven information is quantitative
2. Are my visuals meant to be declarative or exploratory?
 - A declarative purpose is to make a statement
 - An exploratory purpose is to look for new ideas

Four Types of Data Visualizations



Four Types of Data Visualizations



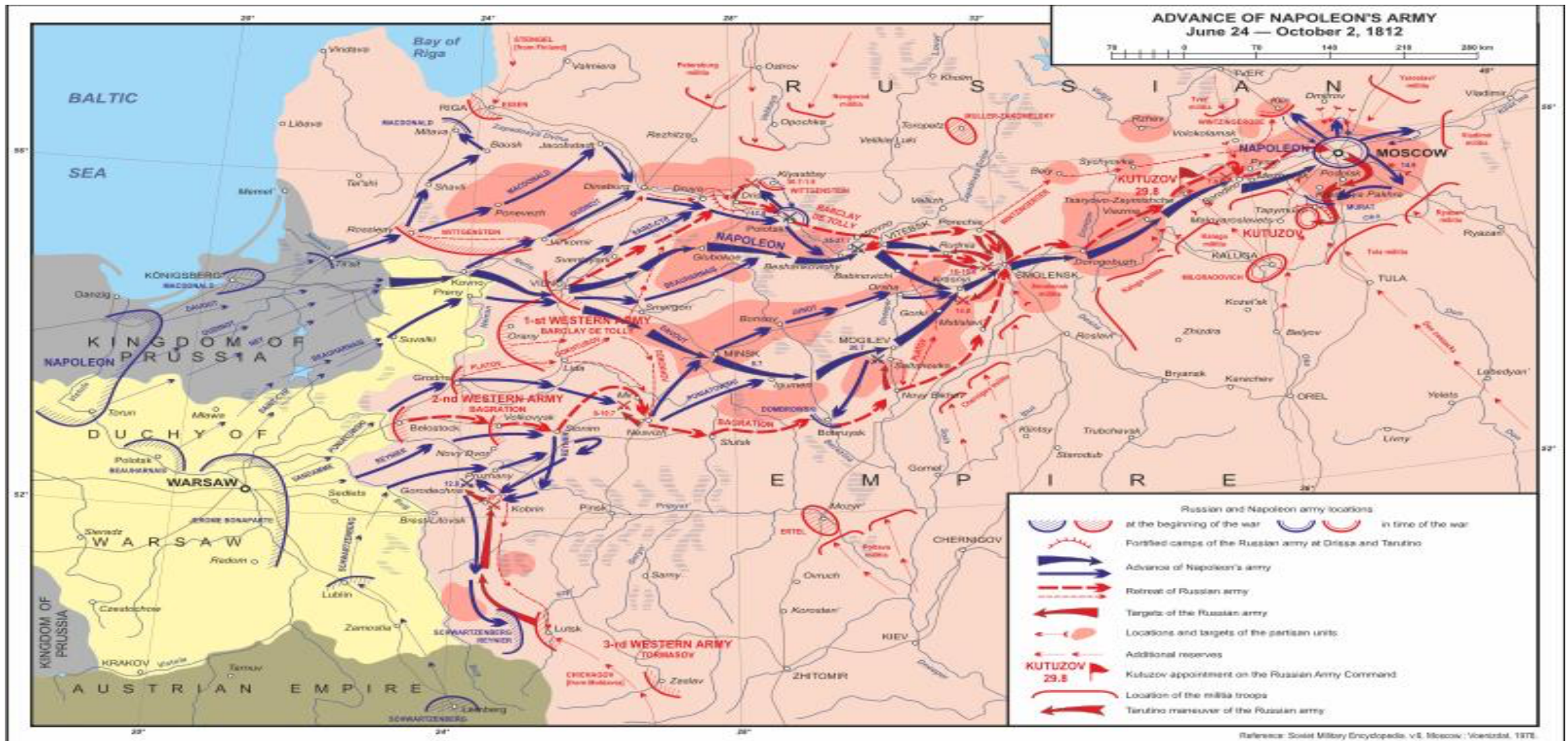
Data Visualization

provide clear understanding of patterns in data

detect hidden structures in data

condense information

What makes a good chart?



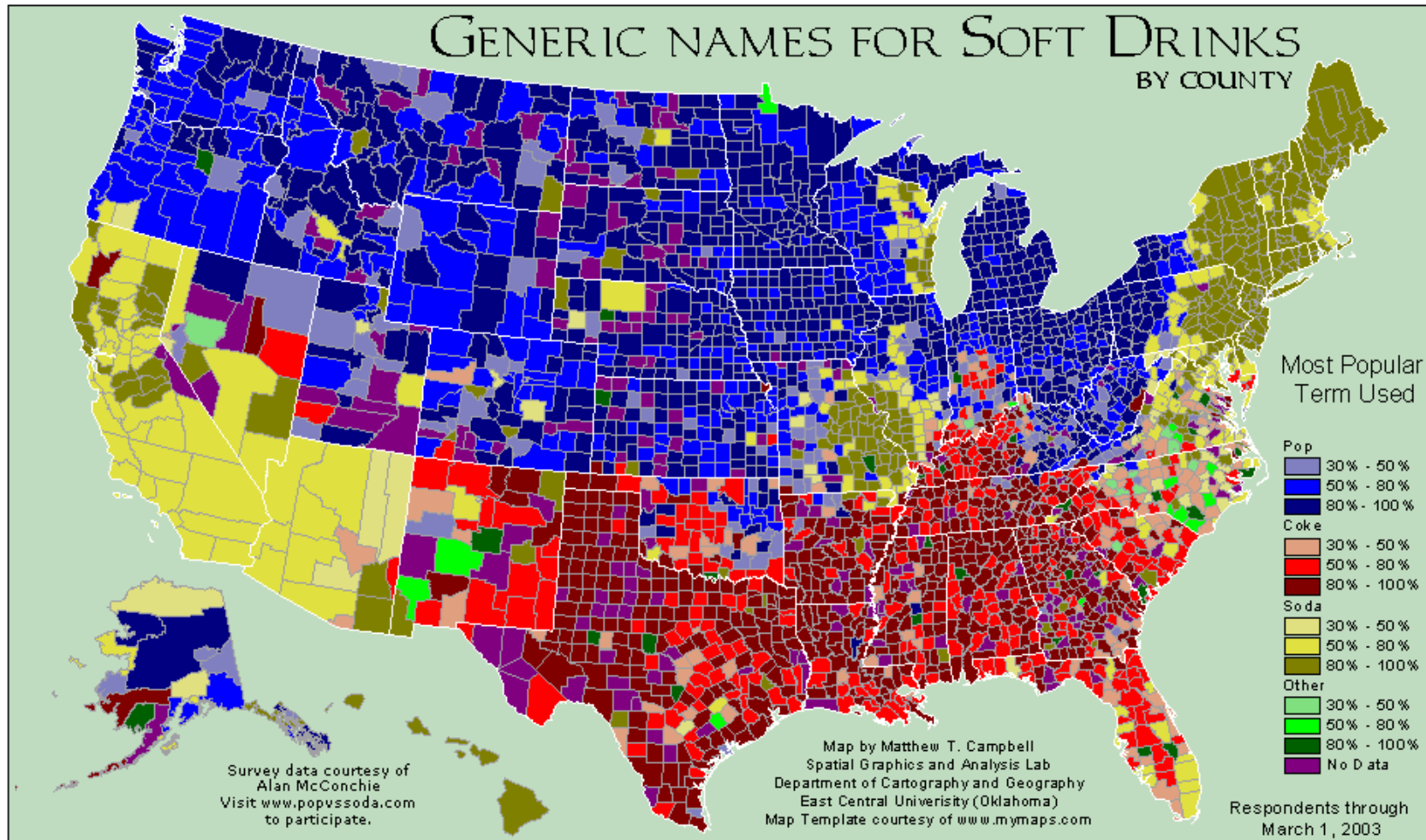
Wikipedia: Patriotic War of 1812

[Video: Napoleonic Wars in 8 Minutes](#)

[Another video](#)

http://en.wikipedia.org/wiki/File:Patriotic_War_of_1812_ENG_map1.svg

What can you learn from this map?



Some basic principles (adapted from Tufte 2009)

1

- The chart should tell a story

2

- The chart should have graphical integrity

3

- The chart should minimize graphical complexity

Tufte's fundamental principle:
Above all else show the data

Principle 1: The chart should tell a story

Graphics should be clear on their own

The depictions should enable meaningful comparison

The chart should yield insight beyond the text

“If the statistics are boring, then you’ve got the wrong numbers.” (Tufte 2009)

Principle 2: The chart should have graphical integrity

- Basically, it shouldn't "lie" (mislead the reader)
- Tufte's "Lie Factor":
 - $Lie\ Factor = \frac{\text{size of effect shown in graphic}}{\text{size of effect in data}}$

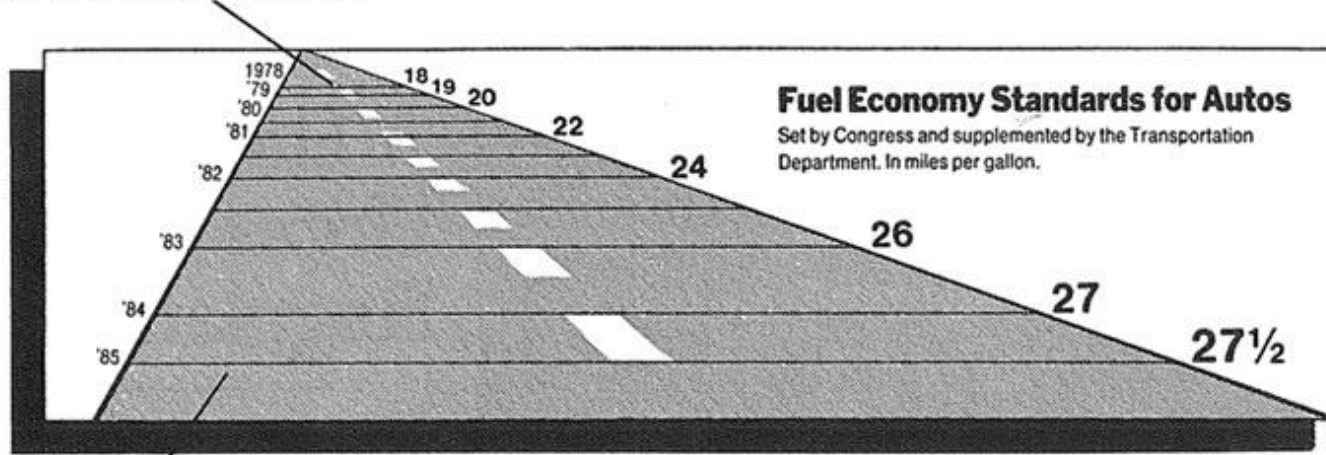
Should be ~ 1

> 1 = exaggerated
effect

< 1 = understated
effect

Examples of the “lie factor”

This line, representing 18 miles per gallon in 1978, is 0.6 inches long.



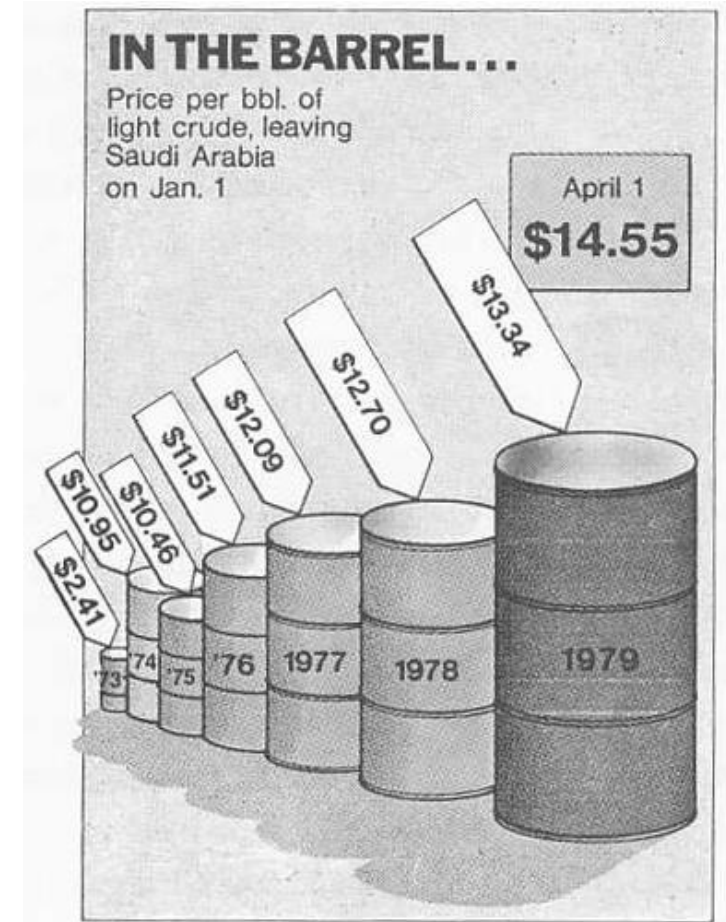
This line, representing 27.5 miles per gallon in 1985, is 5.3 inches long.



$$LF = \frac{5.3/0.6}{27.5/18} = \frac{8.83}{1.53} = 5.77$$

Reprinted from
Tufte (2009), p.
57 & p. 62

$$LF = \frac{4280\% \text{ (change in volume)}}{454\% \text{ (change in price)}} = 9.4 \rightarrow$$



Principle 3: The chart should minimize graphical complexity

Generally, the simpler the better...

Key concepts

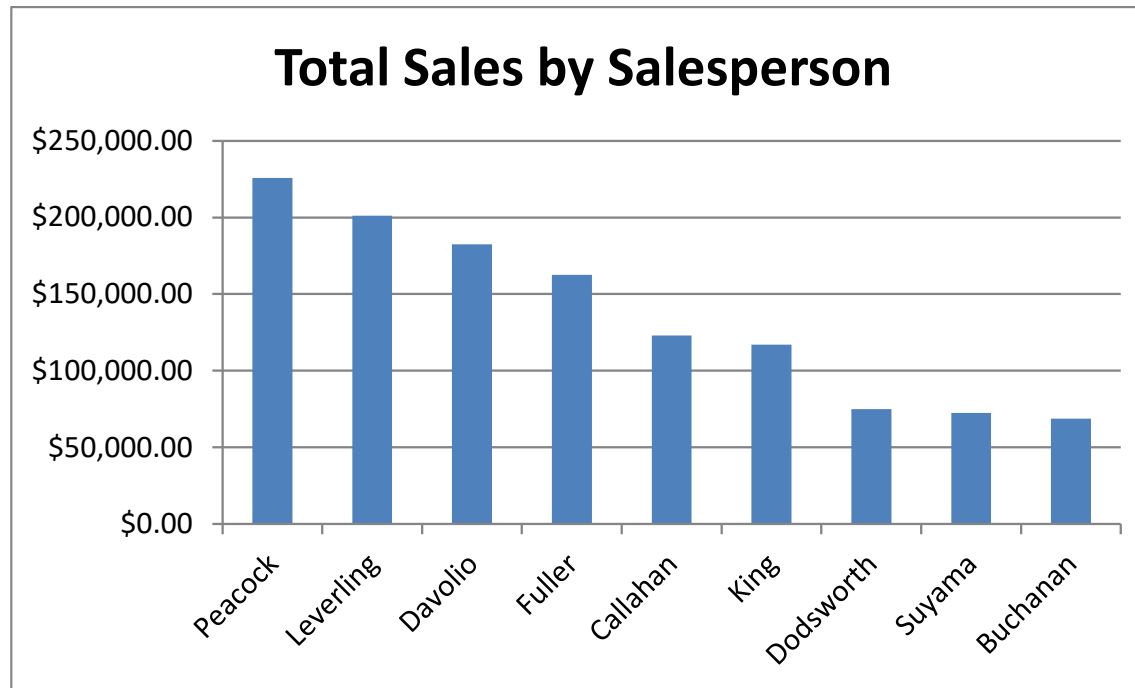
Sometimes
a table is
better

Data-ink

Chart junk

When a table is better than a chart

For a few data points, a table can do just as well...



Salesperson	Total Sales
Peacock	\$225,763.68
Leverling	\$201,196.27
Davolio	\$182,500.09
Fuller	\$162,503.78
Callahan	\$123,032.67
King	\$116,962.99
Dodsworth	\$75,048.04
Suyama	\$72,527.63
Buchanan	\$68,792.25

The table carries more information in less space
and is more precise.

The Ultimate Table: The Box Score

- Large amount of information in a very small space
- So why does this work?
 - Depends on the reader's knowledge of the data



Philadelphia Phillies

Hitters	AB	R	H	RBI	BB	SO	#P	AVG	OBP	SLG
S Victorino CF	3	0	0	0	1	0	16	.000	.250	.000
P Polanco 3B	3	1	0	0	1	0	18	.000	.250	.000
J Rollins SS	4	2	2	0	0	0	14	.500	.500	.500
R Howard 1B	3	1	2	1	0	0	15	.667	.500	.667
R Ibanez LF	4	0	0	1	0	0	14	.000	.000	.000
B Francisco RF	3	1	1	1	1	0	17	.333	.500	.333
C Ruiz C	4	0	1	0	0	0	16	.250	.250	.250
W Valdez 2B	4	0	2	1	0	0	7	.500	.500	.750
R Halladay P	1	0	0	0	0	0	2	.000	.000	.000
a- P Orr PH	1	0	0	0	0	0	3	.000	.000	.000
J Romero P	0	0	0	0	0	0	0	.000	.000	.000
D Herndon P	0	0	0	0	0	0	0	.000	.000	.000
R Madson P	0	0	0	0	0	0	0	.000	.000	.000
b- R Gload PH	1	0	1	0	0	0	3	1.000	1.000	1.000
D Baez P	0	0	0	0	0	0	0	.000	.000	.000
c- J Mayberry Jr. PH	1	0	1	1	0	0	5	1.000	1.000	1.000
Totals	32	5	10	5	3	0	130			

a-lined out to first for R Halladay in the 6th

b-singled to left center for R Madson in the 8th

c-singled to deep center for D Baez in the 9th

Data Ink

- The amount of “ink” devoted to data in a chart
- Tufte’s Data-Ink ratio:

- $$\text{Data – ink ratio} = \frac{\text{data-ink}}{\text{total ink used in graphic}}$$

Should be ~ 1

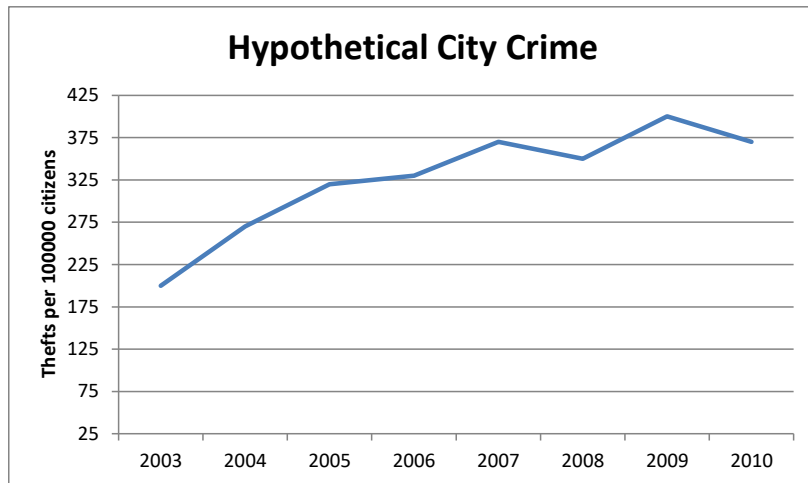
< 1 = more non-data
related ink in graphic

$= 1$ implies all ink
devoted to data

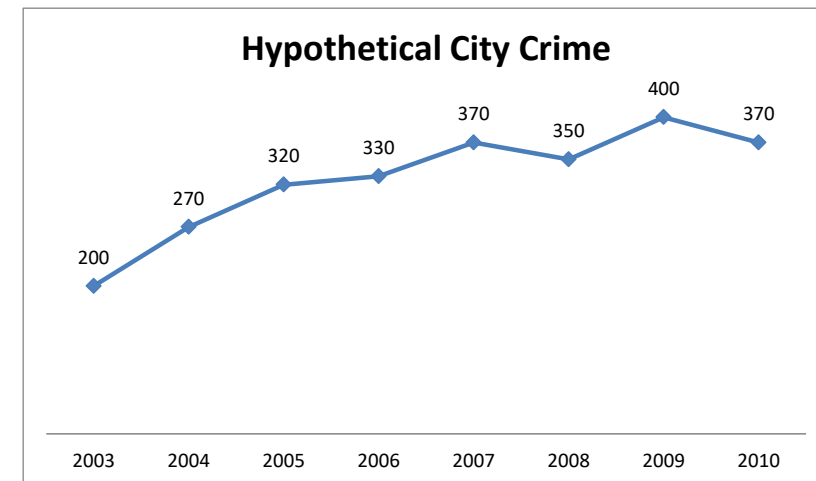
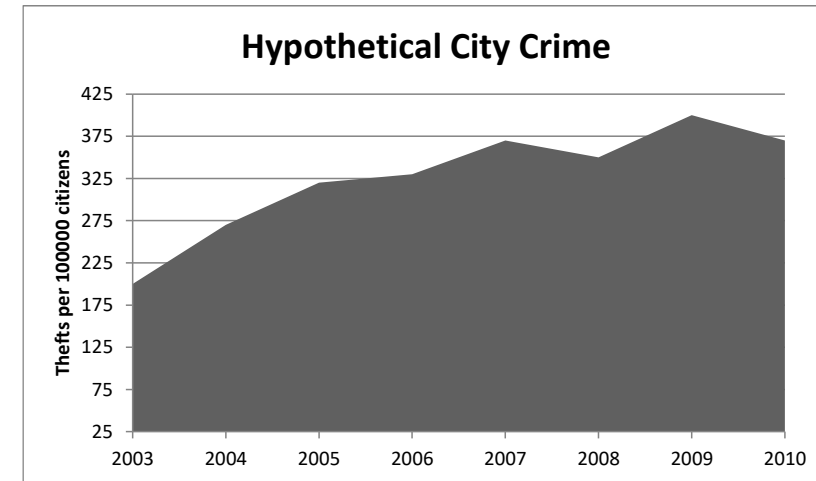
Tufte’s principle:
Erase ink whenever possible

Being conscious of data ink

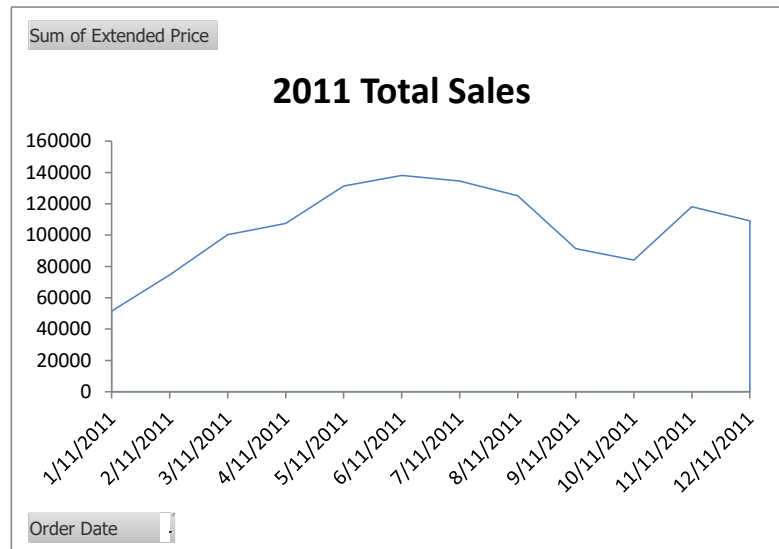
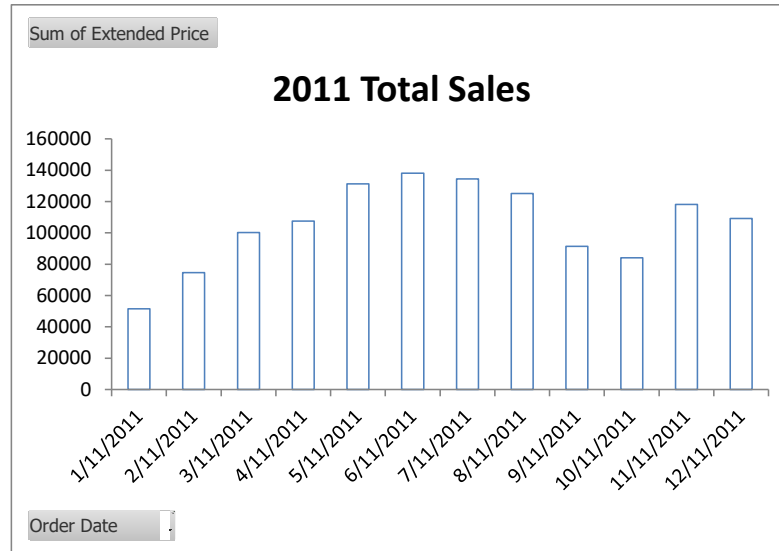
Lower data-ink ratio
(worse)



Higher data-ink ratio
(better)



What makes a good chart?



Sometimes it's
really a matter of
preference.

These both
minimize data ink.

Why isn't a table
better here?

3-D Charts



Evaluate this from a **data-ink** perspective.
How does it affect the clarity of the chart?

One of the golden rules of data visualization is.....

Never use 3D!

Data Integrity/
Lie Factor

- 3D skews numbers, making them difficult to interpret or compare

Graphical
Complexity

- Adding 3D to graphs introduces unnecessary chart elements like side and floor panels

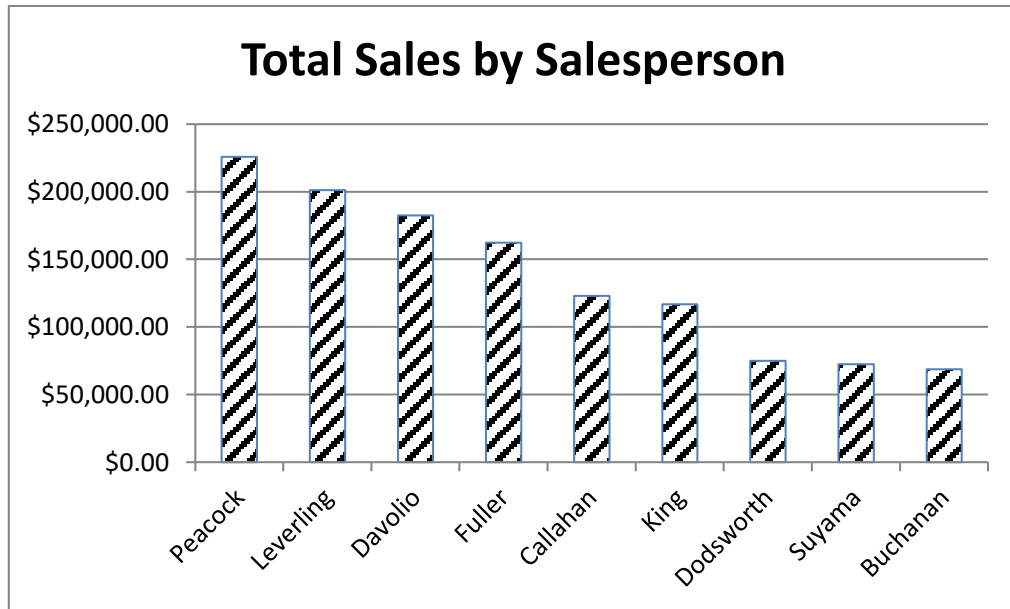
Chartjunk: Data Ink “gone wild”

Unnecessary visual clutter that doesn't provide additional insight

Distraction from the story the chart is supposed to convey

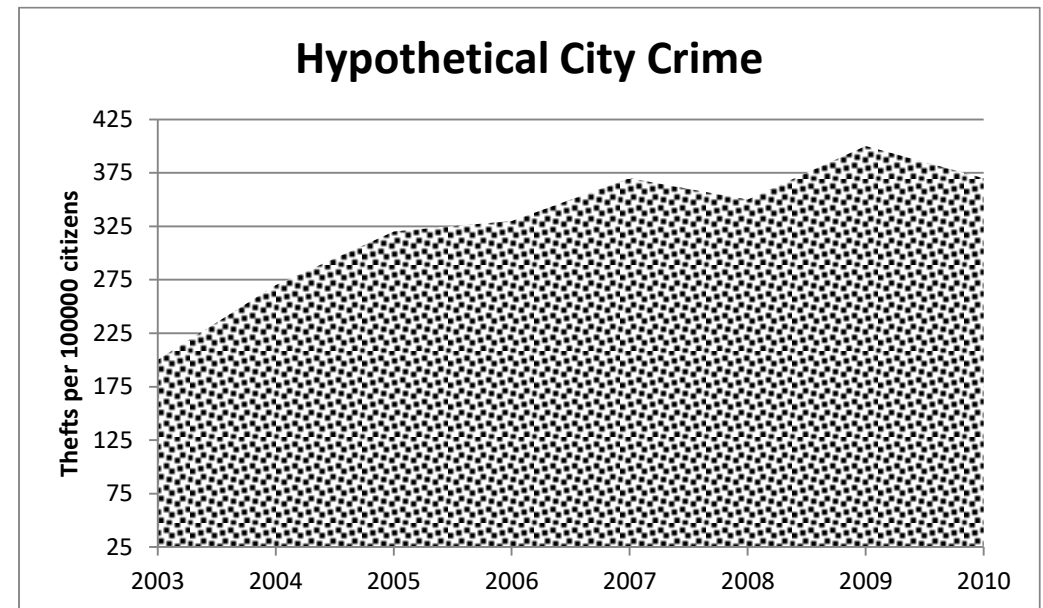
When the data-ink ratio is low, chartjunk is likely to be high

Example: Moiré effects (Tufte 2009)

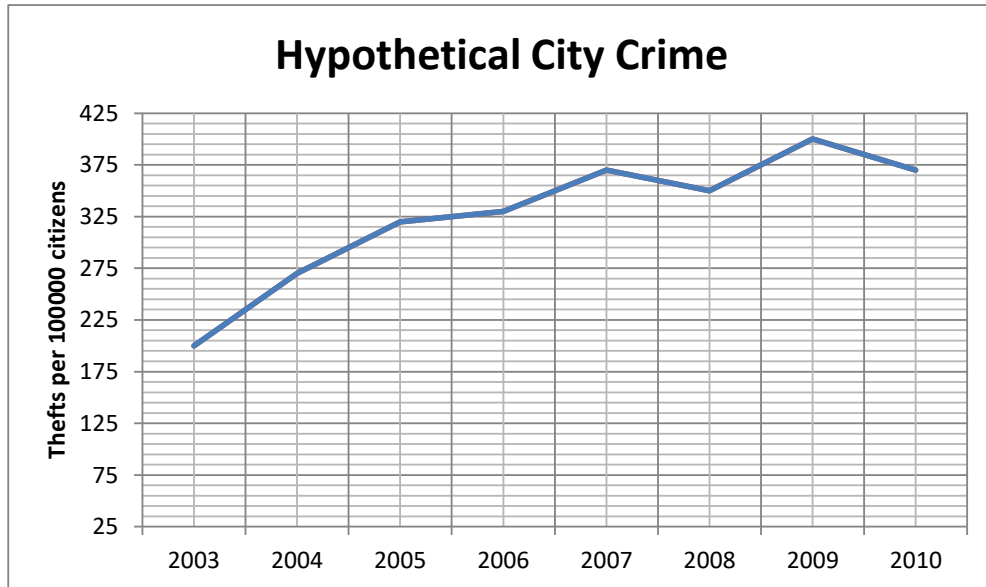


Creates illusion of movement

Stands out, in a bad way

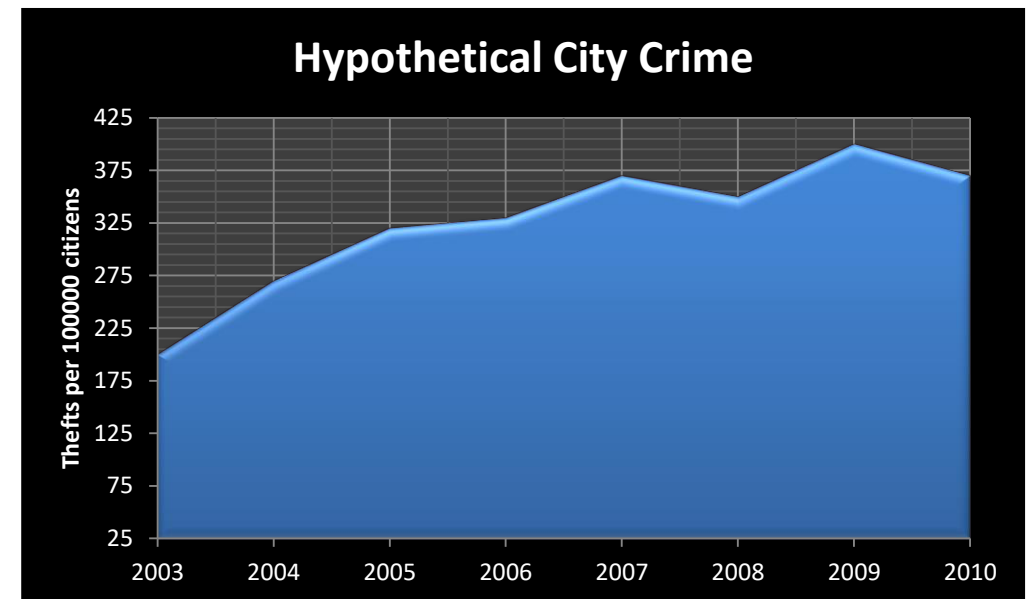


Example: The Grid

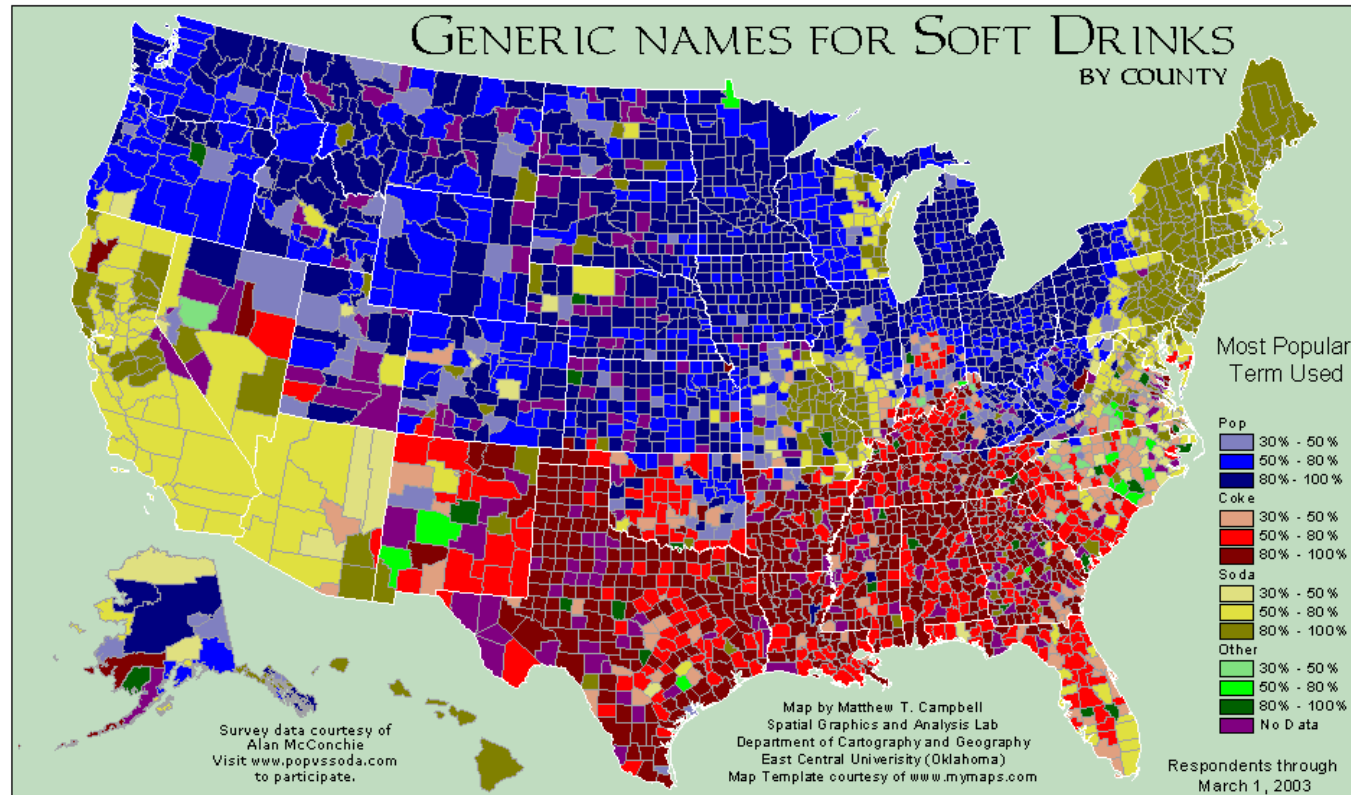


What could you do to remedy it?

Why are these examples of chartjunk?



Data Ink Working For Us



Evaluate this chart in terms of Data Ink.

Imagine this as a bar chart. As a table!!

Review: Data principles (adapted from Tufte 2009)

1

- The chart should tell a story

2

- The chart should have graphical integrity

3

- The chart should minimize graphical complexity

Tufte's fundamental principle:

Above all else show the data

Summary

- Use data visualization principles to assess a visualization
 - Tell a story
 - Graphical integrity (lie factor)
 - Minimize graphical complexity (data ink, chartjunk)
- Explain how a visualization can be improved based on those principles
- Types of visualization

Resources...

- [DataMed](https://datamed.org/) <https://datamed.org/>
- [Institute for Health Metrics and Evaluation's Global Health Data Exchange](http://ghdx.healthdata.org/)
<http://ghdx.healthdata.org/>
- [NNLM RD3: Resources for Data-Driven Discovery](https://nnlm.gov/data/) <https://nnlm.gov/data/>
- [NNLM's YouTube Channel](https://www.youtube.com/channel/UCmZqoegBFKJQF69V8d-05Bw) <https://www.youtube.com/channel/UCmZqoegBFKJQF69V8d-05Bw>
- [OHSU's Big Data to Knowledge](https://dmice.ohsu.edu/bd2k/topics.html) <https://dmice.ohsu.edu/bd2k/topics.html>
- [Registry of Research Data Repositories \(re3data.org\)](http://www.re3data.org/) <http://www.re3data.org/>

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THANK YOU

